and

$$\lambda_\tau(\tau^{-1}\rho) = \tau(\tau^{-1}\rho) = \rho.$$

Thus λ_τ maps onto B_n. Hence the number of elements in A_n is the same as the number in B_n since there is a one-to-one correspondence between the elements of the sets.

Note that the product of two even permutations is again even. Also since $n \geq 2$, S_n has the transposition $(1, 2)$ and $\iota = (1, 2)(1, 2)$ is an even permutation. Finally, note that if σ is expressed as a product of transpositions, the product of the same transpositions taken in just the opposite order is σ^{-1}. Thus if σ is an even permutation, σ^{-1} must also be even. Referring to Theorem 5.12, we see that we have proved the following statement.

8.24 Theorem If $n \geq 2$, then the collection of all even permutations of $\{1, 2, 3, \cdots, n\}$ forms a subgroup of order $n!/2$ of the symmetric group S_n.

We can define a function called the **sign of a permutation**, $\text{sgn} : S_n \to \{1, -1\}$ by the formula

$$\text{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Thinking of $\{1, -1\}$ as a group under multiplication, it is easy to see that sgn is a homomorphism. Since 1 is the identity in the group $\{1, -1\}$, $\text{Ker}(\text{sgn}) = \text{sgn}^{-1}[\{1\}]$ is a subgroup of S_n consisting of all the even permutations. The homomorphism sgn is used in the standard way of defining the determinant of a square matrix. Exercise 52 asks you to prove some of the standard facts about determinants using this definition.

8.25 Definition The subgroup of S_n consisting of the even permutations of n letters is the **alternating group A_n on n letters**. ■

Both S_n and A_n are very important groups. Cayley's theorem shows that every finite group G is structurally identical to some subgroup of S_n for $n = |G|$. It can be shown that there are no formulas involving just radicals for solution of polynomial equations of degree n for $n \geq 5$. This fact is actually due to the structure of A_n, surprising as that may seem!

EXERCISES 8

Computations

In Exercises 1 through 10 determine whether the given map is a group homomorphism. [Hint: To verify that a map is a homomorphism, you must check the homomorphism property. To check that a map is not a homomorphism you could either find a and b such that $\phi(ab) \neq \phi(a)\phi(b)$, or else you could determine that any of the properties in Theorem 8.5 fail.]

1. Let $\phi : \mathbb{Z}_{10} \to \mathbb{Z}_2$ be given by $\phi(x) =$ the remainder when x is divided by 2.
2. Let $\phi : \mathbb{Z}_9 \to \mathbb{Z}_2$ be given by $\phi(x) =$ the remainder when x is divided by 2.
3. Let $\phi : \mathbb{Q}^* \to \mathbb{Q}^*$ be given by $\phi(x) = |x|$.
4. Let $\phi : \mathbb{R} \to \mathbb{R}^+$ be given by $\phi(x) = 2^x$.
5. Let $\phi : D_4 \to \mathbb{Z}_4$ be given by $\phi(\rho^i) = \phi(\mu\rho^i) = i$ for $0 \leq i \leq 3$.

6. Let F be the additive group of all functions mapping $\mathbb{R}$ to $\mathbb{R}$. Let $\phi : F \to F$ be given by $\phi(f) = g$ where $g(x) = f(x) + x$.

7. Let F be as in Exercise 6 and $\phi : F \to F$ be defined by $\phi(f) = 5f$.

8. Let F be the additive group of all continuous functions mapping $\mathbb{R}$ to $\mathbb{R}$. Let $\phi : F \to \mathbb{R}$ be defined by $\phi(g) = \int_0^1 g(x)\,dx$.

9. Let M_n be the additive group of $n \times n$ matrices with real entries. Let $\phi : M_n \to \mathbb{R}$ be given by $\phi(A) = \det(A)$, the determinant of A.

10. Let M_n be as in Exercise 9 and $\phi : M_n \to \mathbb{R}$ be defined by $\phi(A) = \text{tr}(A)$ where $\text{tr}(A)$ is the trace of A, which is the sum of the entries on the diagonal.

In Exercises 11 through 16, compute the kernel for the given homomorphism ϕ.

11. $\phi : \mathbb{Z} \to \mathbb{Z}_8$ such that $\phi(1) = 6$.

12. $\phi : \mathbb{Z} \to \mathbb{Z}$ such that $\phi(1) = 12$.

13. $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\phi(1, 0) = 3$ and $\phi(0, 1) = -5$.

14. $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\phi(1, 0) = 6$ and $\phi(0, 1) = 9$.

15. $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ where $\phi(1, 0) = (2, 5)$ and $\phi(0, 1) = (-3, 2)$.

16. Let D be the additive group of all differentiable functions mapping $\mathbb{R}$ to $\mathbb{R}$ and F the additive group of all functions from $\mathbb{R}$ to $\mathbb{R}$. $\phi : D \to F$ is given by $\phi(f) = f'$, the derivative of f.

In Exercises 17 through 22, find all orbits of the given permutation.

17. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 3 & 6 & 2 & 4 \end{pmatrix}$

18. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 2 & 4 & 8 & 3 & 1 & 7 \end{pmatrix}$

19. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 5 & 1 & 4 & 6 & 8 & 7 \end{pmatrix}$

20. $\sigma : \mathbb{Z} \to \mathbb{Z}$ where $\sigma(n) = n + 1$

21. $\sigma : \mathbb{Z} \to \mathbb{Z}$ where $\sigma(n) = n + 2$

22. $\sigma : \mathbb{Z} \to \mathbb{Z}$ where $\sigma(n) = n - 3$

In Exercises 23 through 25, express the permutation of {1, 2, 3, 4, 5, 6, 7, 8} as a product of disjoint cycles, and then as a product of transpositions.

23. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 2 & 6 & 3 & 7 & 4 & 5 & 1 \end{pmatrix}$

24. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 6 & 4 & 1 & 8 & 2 & 5 & 7 \end{pmatrix}$

25. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 3 & 2 & 8 & 4 & 7 & 6 & 1 \end{pmatrix}$

26. Figure 8.26 shows a Cayley digraph for the alternating group A_4 using the generating set $S = \{(1, 2, 3), (1, 2)(3, 4)\}$. Continue labeling the other nine vertices with the elements of A_4, expressed as a product of disjoint cycles.

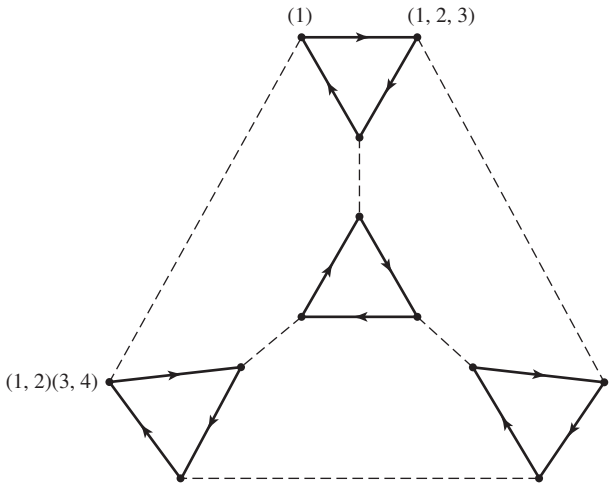
Concepts

In Exercises 27 through 29, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

27. For a permutation σ of a set A, an *orbit* of σ is a nonempty minimal subset of A that is mapped onto itself by σ.

28. The left regular representation of a group G is the map of G into S_G whose value at $g \in G$ is the permutation of G that carries each $x \in G$ into gx.

29. The *alternating group* is the group of all even permutations.



8.26 Figure

30. Before the proof of Cayley's Theorem, it is shown that λ_x is one-to-one. In the proof, one-to-one is shown again. Is it necessary to show one-to-one twice? Explain.

31. Determine whether each of the following is true or false.

a. Every permutation is a cycle.
b. Every cycle is a permutation.
c. The definition of even and odd permutations could have been given equally well before Theorem 8.19.
d. Every nontrivial subgroup H of S_9 containing some odd permutation contains a transposition.
e. A_5 has 120 elements.
f. S_n is not cyclic for any $n \geq 1$.
g. A_3 is a commutative group.
h. S_7 is isomorphic to the subgroup of all those elements of S_8 that leave the number 8 fixed.
i. S_7 is isomorphic to the subgroup of all those elements of S_8 that leave the number 5 fixed.
j. The odd permutations in S_8 form a subgroup of S_8.
k. Every group G is isomorphic with a subgroup of S_G.

32. The dihedral group is defined to be permutations with certain properties. Use the usual notation involving μ and ρ for elements in D_n.

a. Identify which elements in D_3 are even. Do the even elements form a cyclic group?
b. Identify which of elements of D_4 are even. Do the even elements form a cyclic group?
c. For which values of n do the even permutations of D_n form a cyclic group?

Proof Synopsis

33. Give a two-sentence synopsis of the proof of Cayley's Theorem.

34. Give a two-sentence synopsis of the proof of Theorem 8.19.

Theory

35. Suppose that $\phi : G \to G'$ is a group homomorphism and $a \in \text{Ker}\phi$. Show that for any $g \in G$, $gag^{-1} \in \text{Ker}\phi$.

36. Prove that a homomorphism $\phi : G \to G'$ is one-to-one if and only if $\text{Ker}(\phi)$ is the trivial subgroup of G.

37. Let $\phi : G \to G'$ be a group homomorphism. Show that $\phi(a) = \phi(b)$ if and only if $a^{-1}b \in \text{Ker}\phi$.

38. Use Exercise 37 to prove that if $\phi : G \to G'$ is a group homomorphism mapping onto G' and G is a finite group, then for any $b, c \in G'$, $|\phi^{-1}[\{b\}]| = |\phi^{-1}[\{c\}]|$. Conclude that if $|G|$ is a prime number, then either ϕ is an isomorphism or else G' is the trivial group.

39. Show that if $\phi : G \to G'$ and $\gamma : G' \to G''$ are group homomorphisms, then $\gamma \circ \phi : G \to G''$ is also a group homomorphism.

40. Let $\phi : G \to G'$ be a group homomorphism. Show that $\phi[G]$ is abelian if and only if $xyx^{-1}y^{-1} \in \text{Ker}(\phi)$ for all $x, y \in G$.

41. Prove the following about S_n if $n \geq 3$.

a. Every permutation in S_n can be written as a product of at most $n - 1$ transpositions.

b. Every permutation in S_n that is not a cycle can be written as a product of at most $n - 2$ transpositions.

c. Every odd permutation in S_n can be written as a product of $2n + 3$ transpositions, and every even permutation as a product of $2n + 8$ transpositions.

42. a. Draw a figure like Fig. 8.20 to illustrate that if i and j are in different orbits of σ and $\sigma(i) = i$, then the number of orbits of $(i, j)\sigma$ is one less than the number of orbits of σ.

b. Repeat part (a) if $\sigma(j) = j$ also.

43. Show that for every subgroup H of S_n for $n \geq 2$, either all the permutations in H are even or exactly half of them are even.

44. Let σ be a permutation of a set A. We shall say "σ **moves** $a \in A$" if $\sigma(a) \neq a$. If A is a finite set, how many elements are moved by a cycle $\sigma \in S_A$ of length n?

45. Let A be an infinite set. Let H be the set of all $\sigma \in S_A$ such that the number of elements moved by σ (see Exercise 44) is finite. Show that H is a subgroup of S_A.

46. Let A be an infinite set. Let K be the set of all $\sigma \in S_A$ that move (see Exercise 44) at most 50 elements of A. Is K a subgroup of S_A? Why?

47. Consider S_n for a fixed $n \geq 2$ and let σ be a fixed odd permutation. Show that every odd permutation in S_n is a product of σ and some permutation in A_n.

48. Show that if σ is a cycle of odd length, then σ^2 is a cycle.

49. Following the line of thought opened by Exercise 48, complete the following with a condition involving n and r so that the resulting statement is a theorem:

If σ is a cycle of length n, then σ^r is also a cycle if and only if . . .

50. Show that S_n is generated by $\{(1, 2), (1, 2, 3, \cdots, n)\}$. [*Hint:* Show that as r varies, $(1, 2, 3, \cdots, n)^r(1, 2)(1, 2, 3, \cdots, n)^{n-r}$ gives all the transpositions $(1, 2), (2, 3), (3, 4), \cdots, (n - 1, n), (n, 1)$. Then show that any transposition is a product of some of these transpositions and use Theorem 8.15.]

51. Let $\sigma \in S_n$ and define a relation on $\{1, 2, 3, \ldots, n\}$ by $i \sim j$ if and only if $j = \sigma^k(i)$ for some $k \in \mathbb{Z}$.

1. Prove that $\sim$ is an equivalence relation.

2. Prove that for any $1 \leq i \leq n$, the equivalence class of i is the orbit of i.

52. The usual definition for the determinant of an $n \times n$ matrix $A = (a_{i,j})$ is

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1,\sigma(1)} a_{2,\sigma(2)} a_{3,\sigma(3)} \cdots a_{n,\sigma(n)}$$

where $\text{sgn}(\sigma)$ is the sign of σ. Using this definition, prove the following properties of determinants.

a. If a row of matrix A has all zero entries, then $\det(A) = 0$.

b. If two different rows of A are switched to obtain B, then $\det(B) = -\det(A)$.

c. If r times one row of A is added to another row of A to obtain a matrix B, then $\det(A) = \det(B)$

d. If a row of A is multiplied by r to obtain the matrix B, then $\det(B) = r \det(A)$.

53. Prove that any finite group G is isomorphic with a subgroup of $\text{GL}(n, \mathbb{R})$ for some n. [Hint: For each $\sigma \in S_n$, find a matrix in $\text{GL}(n, \mathbb{R})$ that sends each basis vector e_i to $e_{\sigma(i)}$. Use this to show that S_n is isomorphic with a subgroup of $\text{GL}(n, \mathbb{R})$.]

54. Prove Cayley's Theorem using the right regular representation rather than the left regular representation.

55. Let $\sigma \in S_n$. An inversion is a pair (i, j) such that $i < j$ and $\sigma(i) > \sigma(j)$. Prove Theorem 8.19 by showing that multiplying a permutation by a transposition changes the number of inversions by an odd number.